

Comparitive Performance Report for tentacle-3-way-replica-balance-reads-off vs tentacle-3-way-replica-balance-reads-on vs direct-reads-3-way-replica-balance-reads- off-pr2-rerun vs direct-reads-3-way-replica-balance-reads- on-pr2-rerun vs direct-reads-3-way-replica-balance-on- pr2-split-256k-rerun

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Comparison summary for
tentacle-3-way-replica-balance-reads-off vs
tentacle-3-way-replica-balance-reads-on vs
direct-reads-3-way-replica-balance-reads-off-pr2-rerun vs
direct-reads-3-way-replica-balance-reads-on-pr2-rerun vs
direct-reads-3-way-replica-balance-on-pr2-split-256k-rerun

Sequential									
Read	tentacle_3	tentacle3-way-replica-balance-reads-off	change	direct-reads-off	reads%3change	replicat	reads%3change	offpr2	reads%3change
4K	157548@2.0ms	161426@1.5	66%	98612@3.9	-37%	143388@2.7	-9%	142860@2.7	-9%
8K	150852@2.5ms	152900@1.5	68%	97433@3.9	-35%	141275@2.7	-6%	141398@2.7	-6%
16K	138108@2.8ms	149887@1.7	59%	94254@4.1	-32%	136141@2.8	-1%	136657@2.8	-1%
32K	121533@3.2ms	129210@2.0	58%	90224@4.3	-26%	127844@3.0	5%	128252@3.0	6%
64K	6331@4.0ms	9299@2.7	47%	5215@4.8	-18%	6921@3.6	9%	7001@3.6	11%
128K	9823@5.1ms	13091@3.8	33%	8829@5.7	-10%	10840@4.6	10%	10982@4.6	12%
256K	14295@7.0ms	17215@5.8	20%	13856@7.3	-3%	14965@6.7	5%	15064@6.7	5%
384K	14995@6.7ms	16934@11.9	13%	14198@7.1	-5%	15340@6.6	2%	15358@13.1	2%
512K	17061@15.7ms	14250@18.8	-16%	15051@8.9	-12%	15641@8.6	-8%	13469@19.9	-21%
768K	17212@23.4ms	16851@11.9	-2%	14516@8.7	-16%	15917@7.9	-8%	13634@9.2	-21%
1024K	16689@24.1ms	15398@17.4	-8%	14482@9.3	-13%	15492@17.3	-7%	14077@28.6	-16%
2048K	14689@9.1ms	14321@28.1	-3%	13073@10.2	-11%	14161@9.5	-4%	14724@4.5	0%
4096K	13597@59.2ms	14110@57.1	4%	12152@9.6	-11%	12605@10.6	-7%	14547@36.9	7%

Random									
Read	tentacle_3	tentacle3-way-replica-balance-reads-off	change	direct-reads-off	reads%3change	replicat	reads%3change	offpr2	reads%3change
4K	219491@1.2ms	239094@1.1	9%	161654@2.4	-26%	154585@2.5	-30%	154009@2.5	-30%
8K	209044@1.2ms	225342@1.7	8%	156857@2.4	-25%	148781@2.6	-29%	150647@2.5	-28%
16K	188007@1.4ms	203290@1.9	8%	151354@2.5	-19%	142960@2.7	-24%	144210@2.7	-23%
32K	167762@2.3ms	182074@1.8	9%	140228@2.7	-16%	134125@2.9	-20%	133819@2.9	-20%
64K	9032@2.8ms	9412@2.7	4%	7552@2.8	-16%	7426@3.4	-18%	7408@3.4	-18%
128K	13244@3.8ms	13212@2.5	-0%	10883@3.1	-18%	10784@4.7	-19%	11272@4.5	-15%
256K	16481@4.1ms	14106@7.1	-14%	14122@2.4	-14%	14012@4.8	-15%	13172@7.6	-20%
384K	17128@11.7ms	16696@3.8	-3%	15356@13.1	-10%	14897@3.4	-13%	14859@4.2	-13%
512K	17600@15.2ms	15960@4.2	-9%	15723@4.3	-11%	15701@4.3	-11%	12874@2.6	-27%
768K	17110@23.5ms	16308@6.2	-5%	16213@6.2	-5%	15642@6.4	-9%	12986@3.9	-24%

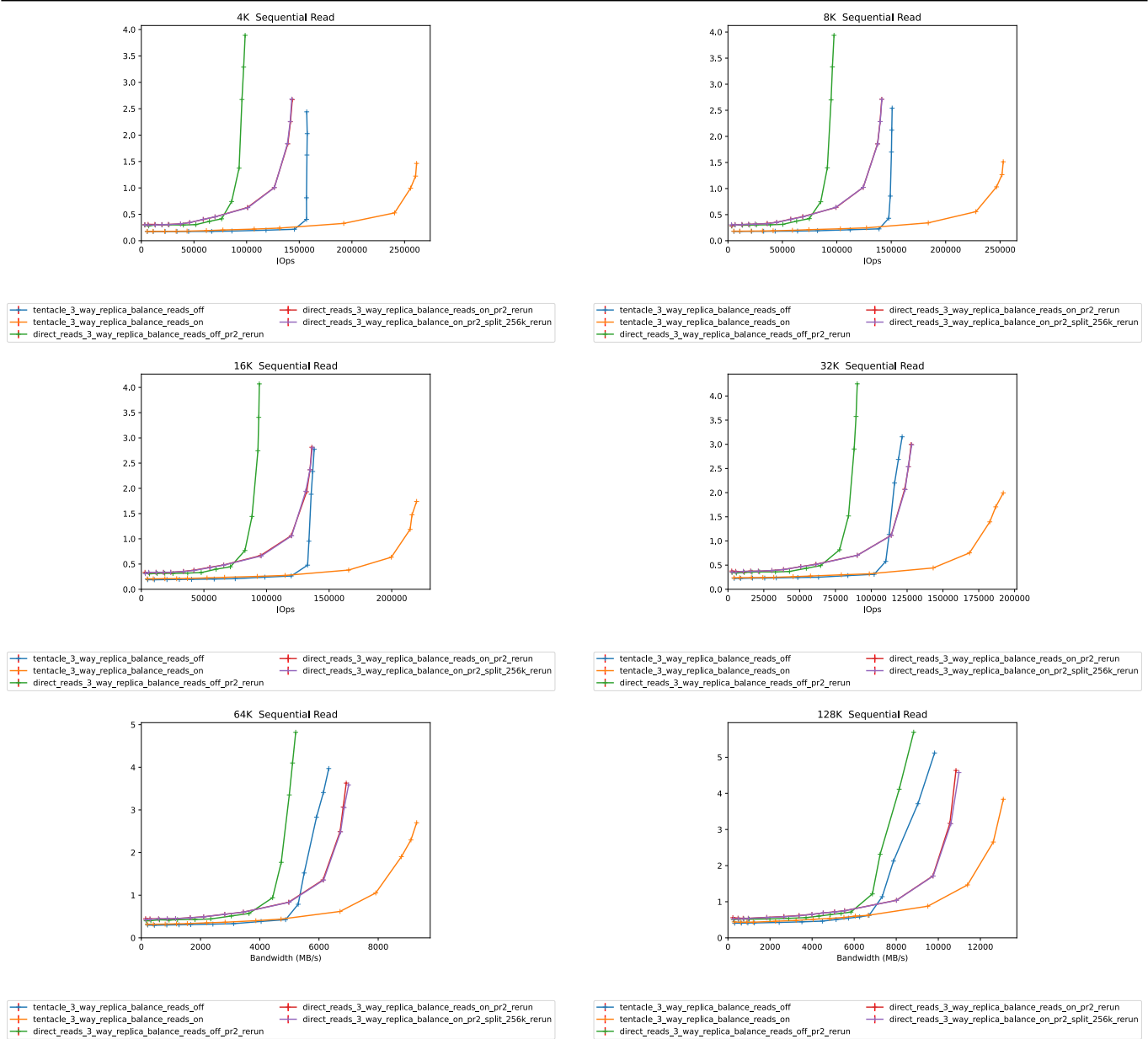
Comparison summary for tentacle-3-way-replica-balance-reads-off vs tentacle-3-way-replica-balance-reads-on vs direct-reads-3-way-replica-balance-reads-off-pr2-rerun vs direct-reads-3-way-replica-balance-reads-on-pr2-rerun vs direct-reads-3-way-replica-balance-on-pr2-split-256k-rerun

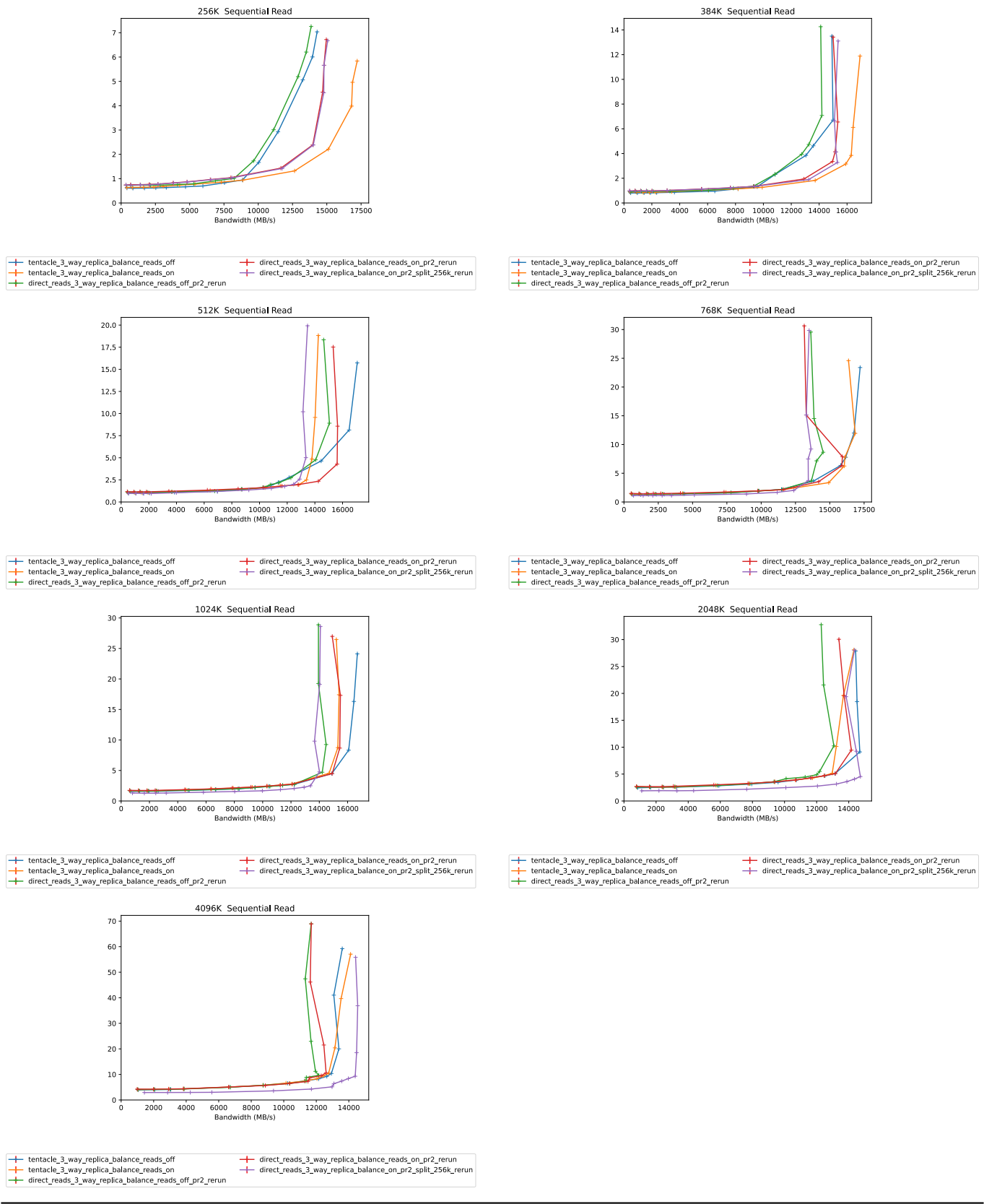
Random									
Read	tentacle_3_way-replica-balance-reads-off	tentacle_3_way-replica-balance-reads-on	%change	direct-reads-3-way-replica-balance-reads-off-pr2-rerun	%change	direct-reads-3-way-replica-balance-reads-on-pr2-rerun	%change	direct-reads-3-way-replica-balance-on-pr2-split-256k-rerun	%change
1024K	16507@16.2m	16095@8.3	-2%	14980@4.5	-9%	14236@4.7	-14%	13467@5.0	-18%
2048K	14724@18.2m	14222@18.8	-3%	13194@10.1	-10%	13490@29.8	-8%	14612@4.6	-1%
4096K	13969@57.6m	13727@58.7	-2%	12637@9.3	-10%	12661@63.6	-9%	14435@18.6	3%

Random									
Read/Write	tentacle_3_way-replica-balance-reads-off	tentacle_3_way-replica-balance-reads-on	%change	direct-reads-3-way-replica-balance-reads-off-pr2-rerun	%change	direct-reads-3-way-replica-balance-reads-on-pr2-rerun	%change	direct-reads-3-way-replica-balance-on-pr2-split-256k-rerun	%change
4K_70/30	119218@3.2ms	113422@3.1	4%	96199@4.0	-19%	96971@4.0	-19%	96501@4.0	-19%
16K_70/30	101882@3.8ms	106175@3.6	4%	84567@3.8	-17%	85670@4.5	-16%	85515@4.5	-16%
64K_30/70	3170@5.3ms	2947@2.8	-7%	2764@9.1	-13%	2769@7.6	-13%	2670@6.3	-16%
64K_70/30	4847@4.3ms	4942@5.1	2%	4200@6.0	-13%	4271@5.9	-12%	4275@5.9	-12%

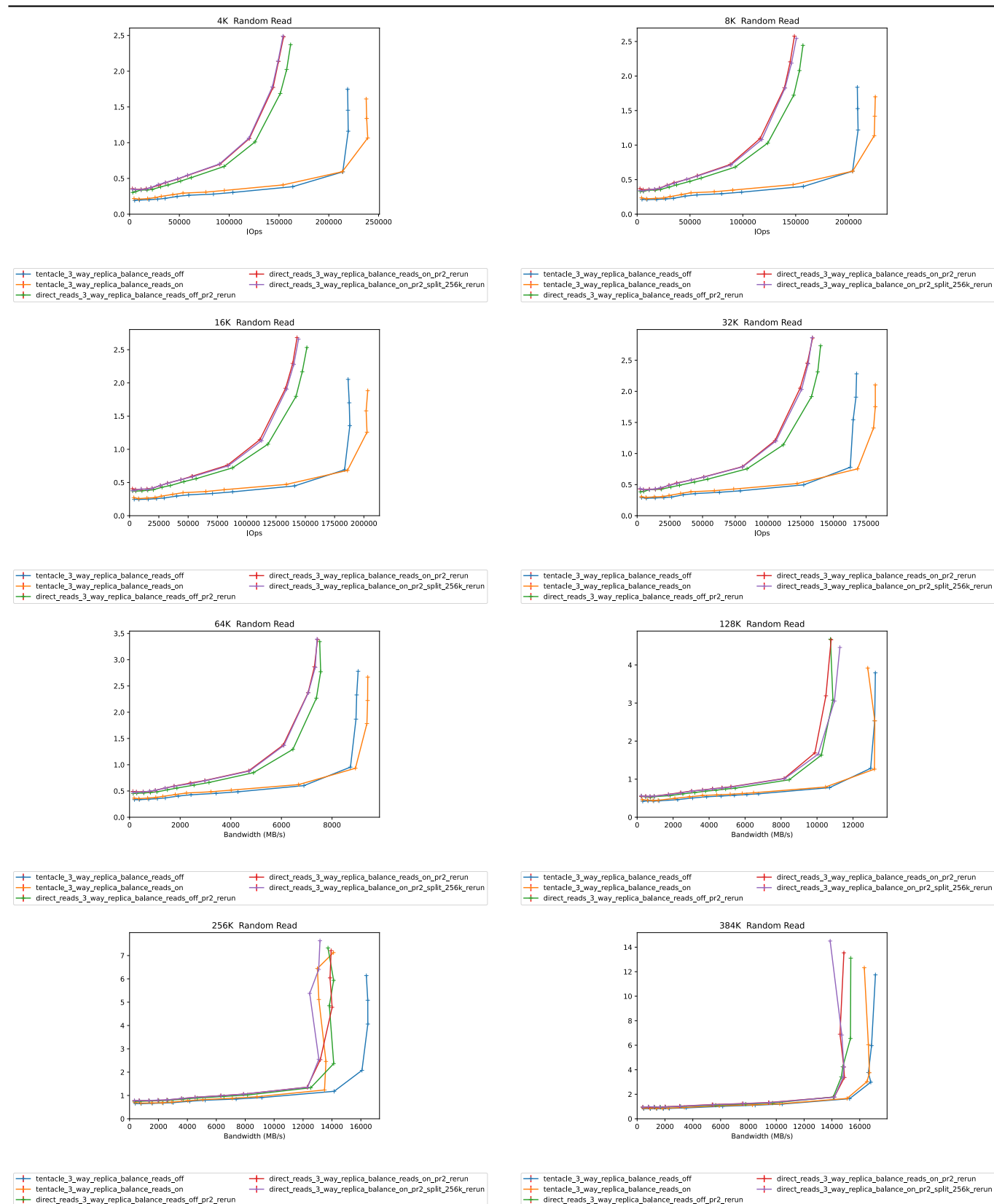
Response Curves

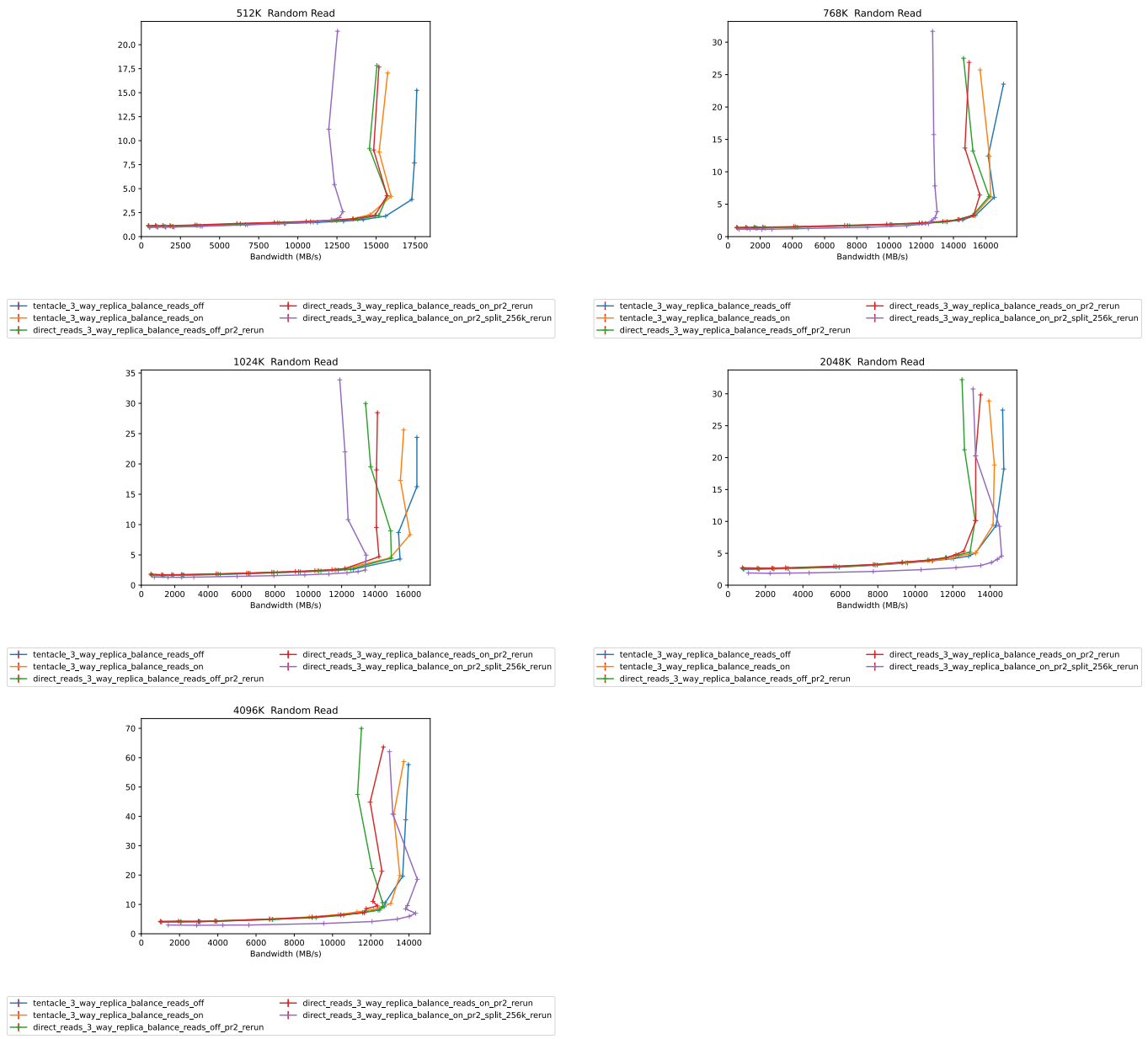
Sequential Read



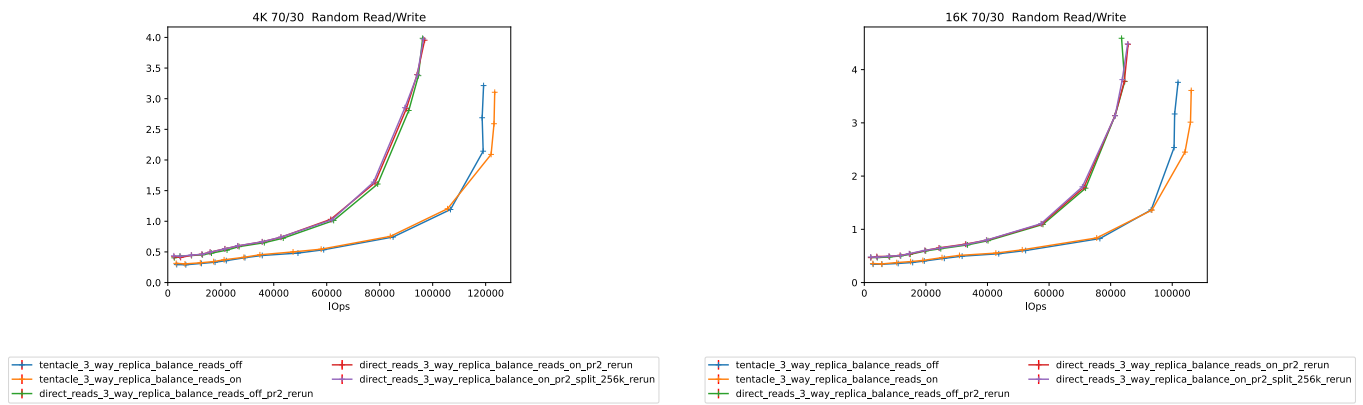


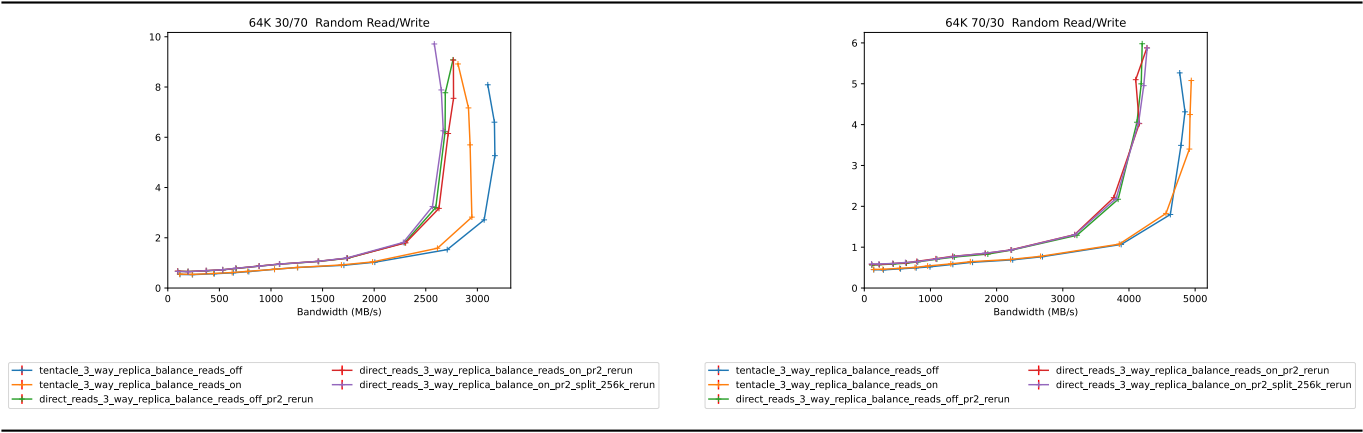
Random Read





Random Read/Write





Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

256ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 262144
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

2Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

2Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```


- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4096
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4096
- total_iodepth:

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

512krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

512ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

64kmixread30:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 65536
- rwmixread: 30

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 65536
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```

    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}

```